

ROAD ACCIDENT SEVERITY PREDICTION SYSTEM

MACHINE LEARNING & TABLEAU DASHBOARD
PROJECT REPORT

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BY

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TECHOLAS
TECHNOLOGIES

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1. INTRODUCTION

Road accidents are a major cause of injuries and fatalities worldwide, resulting from factors such as driver behavior, weather conditions, road conditions, vehicle defects, and traffic violations. Predicting accident severity can help traffic authorities and emergency services improve road safety and take timely preventive measures.

This project develops a Road Accident Severity Prediction System using Machine Learning techniques to classify accident severity based on historical accident data. The dataset includes information about drivers, vehicles, road conditions, weather, and accident characteristics.

The project involves data preprocessing, feature engineering, exploratory data analysis (EDA), model training, hyperparameter tuning, and performance evaluation. Four machine learning algorithms—Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN)—were trained and compared. Among them, Random Forest achieved the highest accuracy and was selected as the final model.

The trained model was deployed using Streamlit, allowing users to upload accident data and receive severity predictions. Additionally, a Tableau dashboard was created to visualize accident trends and provide meaningful insights for better decision-making.

2. ABSTRACT

This project focuses on predicting the severity of road accidents using multiple machine learning classification techniques. The dataset, obtained from Kaggle, contains historical road accident records with information related to driver characteristics, vehicle details, road conditions, weather conditions, collision types, casualty information, and accident causes.

The dataset was preprocessed by handling missing values, performing feature engineering, and encoding categorical variables to prepare it for machine learning. Exploratory Data Analysis (EDA) was conducted to identify accident patterns and understand the relationship between different features and accident severity.

Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN), were trained and evaluated using accuracy and classification metrics. Hyperparameter tuning was applied to the Random Forest classifier using GridSearchCV to improve prediction performance.

The best-performing model was deployed as a Streamlit web application that allows users to upload accident datasets and predict accident severity in real time. Additionally, an interactive Tableau dashboard was developed to visualize accident trends, severity distribution, weather analysis, road conditions, and other key insights. The proposed system demonstrates how machine learning and data visualization can support road safety analysis and assist decision-makers in reducing accident risks.

3. OBJECTIVES

The primary objectives of the Road Accident Severity Prediction System are:

- Develop a machine learning model to predict the severity of road accidents based on historical accident data.
- Perform Exploratory Data Analysis (EDA) to identify accident patterns, trends, and factors influencing accident severity.
- Preprocess the dataset by handling missing values, performing feature engineering, and preparing the data for machine learning.
- Train and evaluate multiple machine learning classification algorithms, including Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN).
- Improve model performance through Hyperparameter Tuning using GridSearchCV.
- Compare the performance of different machine learning models using evaluation metrics such as Accuracy, Precision, Recall, and F1-Score.
- Develop an interactive Tableau dashboard to visualize accident trends, severity distribution, weather conditions, and other important insights.
- Deploy the best-performing machine learning model as a Streamlit web application that enables users to upload accident datasets and obtain real-time accident severity predictions.
- Provide an efficient decision-support system that can assist traffic authorities, researchers, and policymakers in improving road safety and reducing accident-related risks.

4. METHODOLOGY

Step 1: Data Collection

- The dataset was obtained from Kaggle and contains historical road accident records.
- The dataset includes driver information, vehicle details, road conditions, weather conditions, collision types, casualty information, and accident severity.
- The target variable for prediction is Accident Severity.

Step 2: Data Preprocessing

- Loaded the dataset using the Pandas library.
- Checked the dataset structure using `head()`, `info()`, and `describe()`.
- Identified missing values using `isnull().sum()`.
- Filled missing categorical values using the most frequent value (Mode).
- Converted column names into a consistent lowercase format.
- Extracted the Hour feature from the Time column.
- Created a new Time Period feature (Morning, Afternoon, Evening, Night).
- Saved the cleaned dataset for further analysis.

Step 3: Exploratory Data Analysis (EDA)

Several visualizations were created to understand accident patterns and identify important relationships between variables.

The analysis included:

- Accident Severity Distribution
- Driver Age Distribution
- Weather Condition Analysis
- Road Surface Condition Analysis
- Cause of Accident Analysis
- Vehicle Type Distribution

- Day-wise Accident Analysis
- Light Condition Analysis

Step 4: Model Building

The dataset was divided into Training (80%) and Testing (20%) sets. Categorical variables were transformed using One-Hot Encoding through a preprocessing pipeline.

The following machine learning algorithms were trained:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- K-Nearest Neighbors (KNN)

Step 5: Hyperparameter Tuning

Random Forest was selected as the best-performing algorithm after initial model comparison.

Hyperparameter tuning was performed using GridSearchCV to identify the optimal values for:

- Number of Trees (n_estimators)
- Maximum Tree Depth (max_depth)
- Minimum Samples Split (min_samples_split)
- Minimum Samples per Leaf (min_samples_leaf)

The tuned Random Forest model produced the highest prediction accuracy.

Step 6: Model Evaluation

The performance of each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Classification Report

The trained models were compared, and Random Forest achieved the best overall performance.

- Step 7: Tableau Dashboard Development
- An interactive Tableau dashboard was developed to visualize accident statistics and important trends.
- The dashboard includes:
 - Total Accidents
 - Accident Severity Distribution
 - Accidents by Weather Condition
 - Driver Age Distribution
 - Road Surface Analysis
 - Cause of Accident
 - Vehicle Type Analysis
 - Day-wise Accident Distribution
- Step 8: Streamlit Web Application
- The best-performing Random Forest model was saved using Joblib and deployed as a Streamlit web application.
- The application allows users to:
 - Upload CSV, XLSX, XLS, or JSON files.
 - Preview uploaded accident data.
 - Predict accident severity using the trained model.
 - Download the prediction results as a CSV file.
- This deployment provides an easy-to-use interface for real-time accident severity prediction without requiring programming knowledge.

5. DATASET OVERVIEW

The Road Accident Severity Dataset used in this project was obtained from Kaggle. The dataset contains historical records of road accidents along with information related to drivers, vehicles, road infrastructure, weather conditions, collision types, and casualty details. The target variable of the dataset is Accident Severity, which represents the level of injury caused by each accident.

The dataset consists of 12,316 records and 32 attributes before preprocessing. These attributes provide valuable information for analyzing accident patterns and predicting accident severity using machine learning techniques.

The dataset contains both categorical and numerical features. Categorical features include variables such as driver age group, type of vehicle, weather conditions, road surface type, and cause of accident. Numerical features include the number of vehicles involved and the number of casualties. During preprocessing, additional features such as Hour and Time Period were created from the original time column to improve prediction performance.

The target variable Accident Severity consists of three categories:

- Slight Injury
- Serious Injury
- Fatal Injury

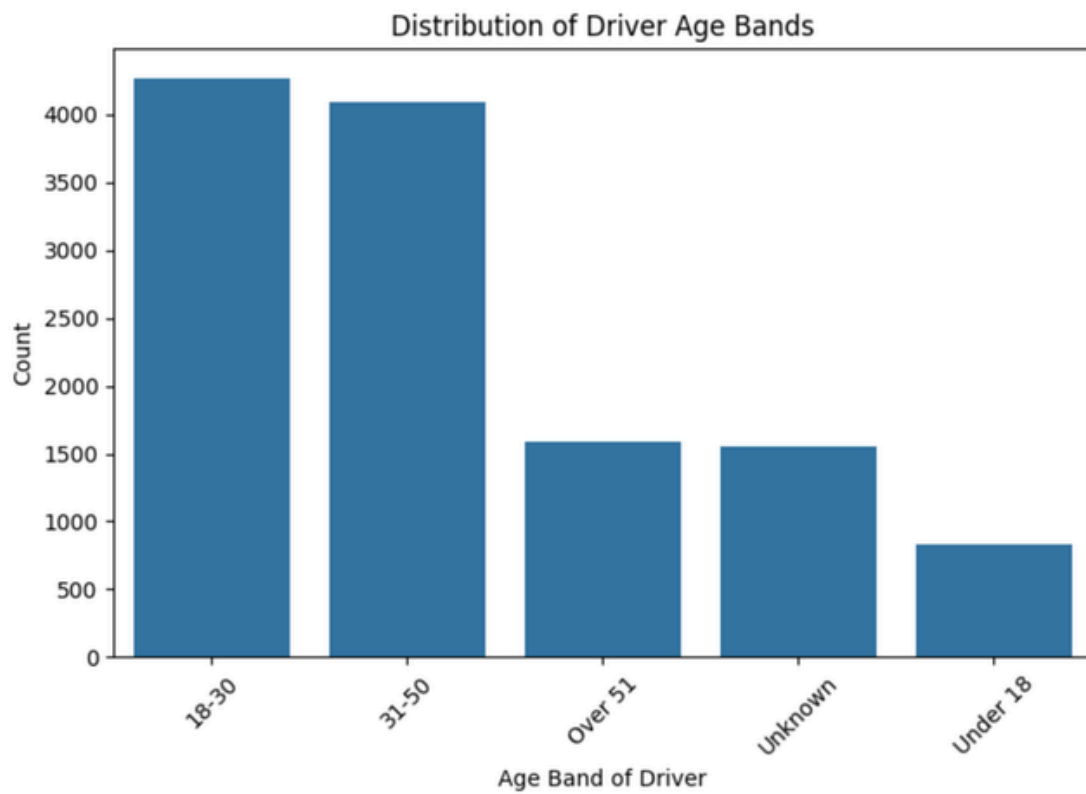
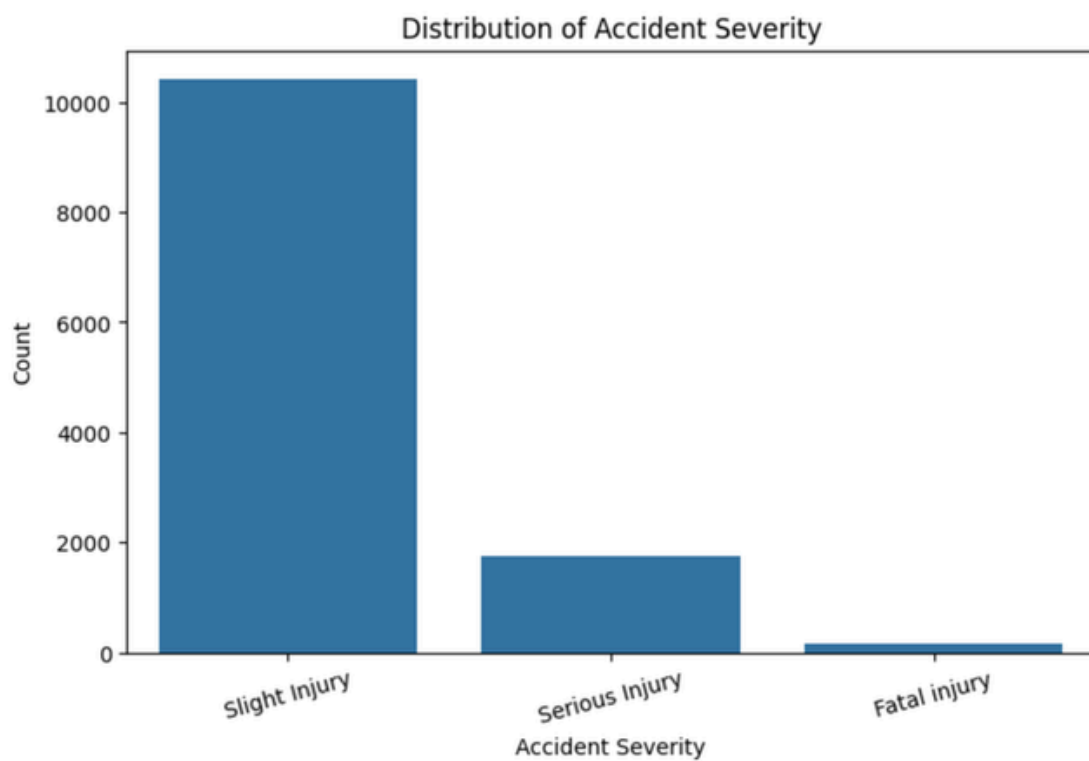
This dataset provides sufficient information to build an effective machine learning model capable of predicting accident severity based on various accident-related factors.

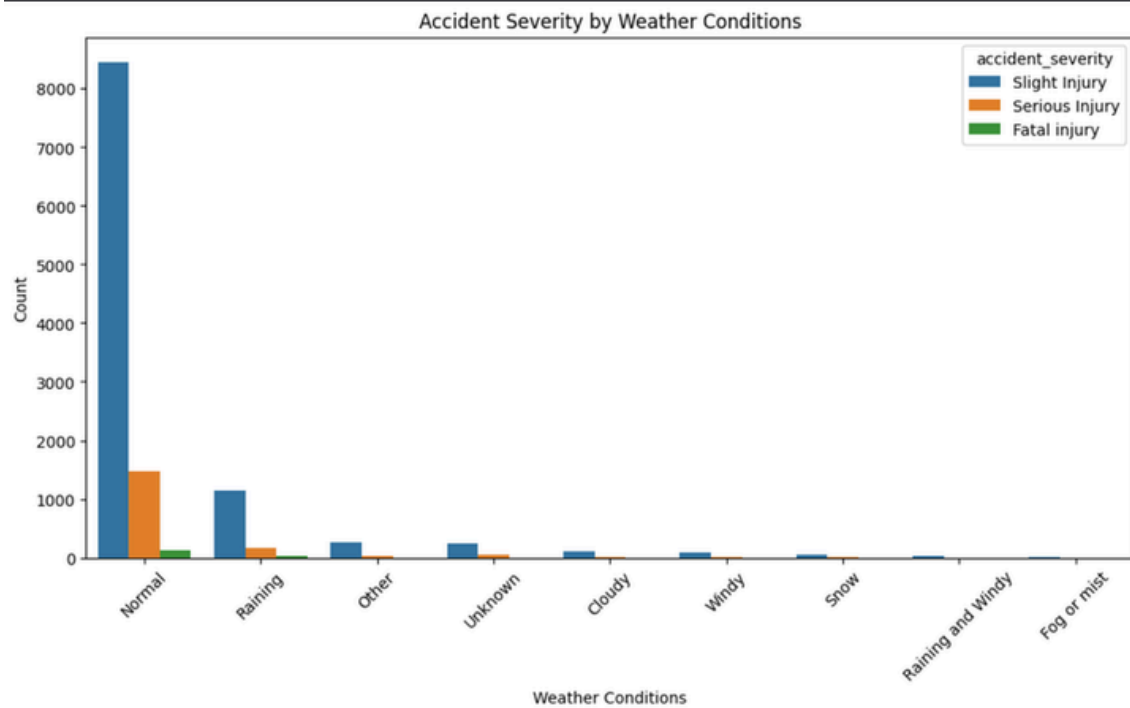
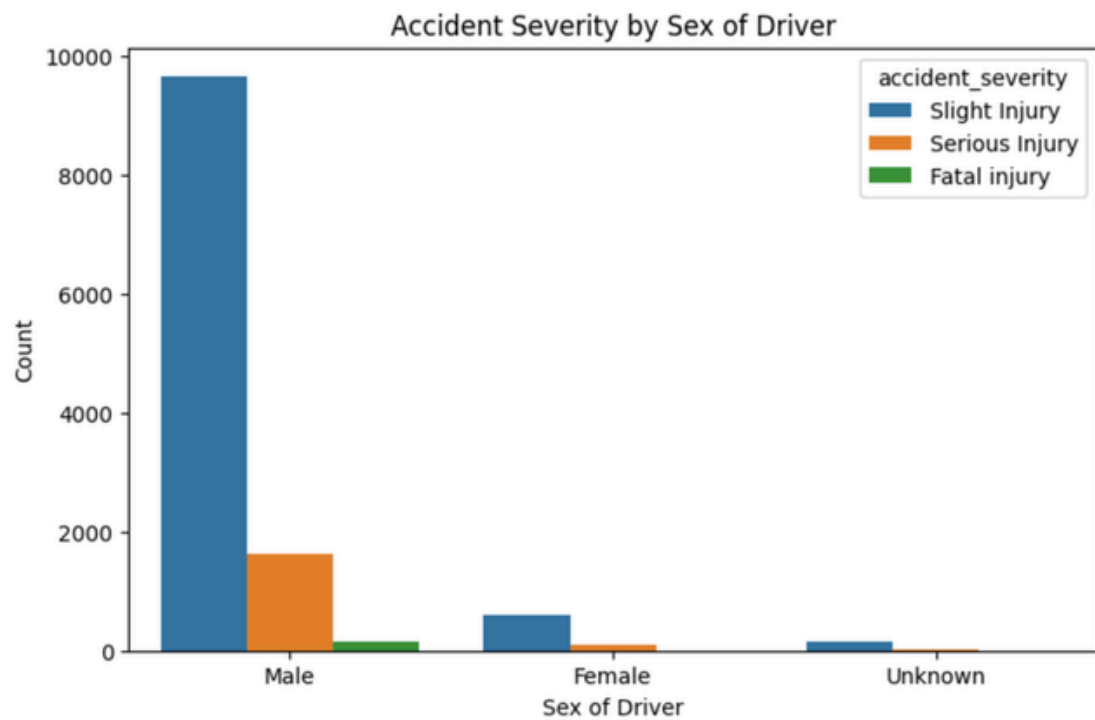
6. DATA ANALYSIS & EDA VISUALS

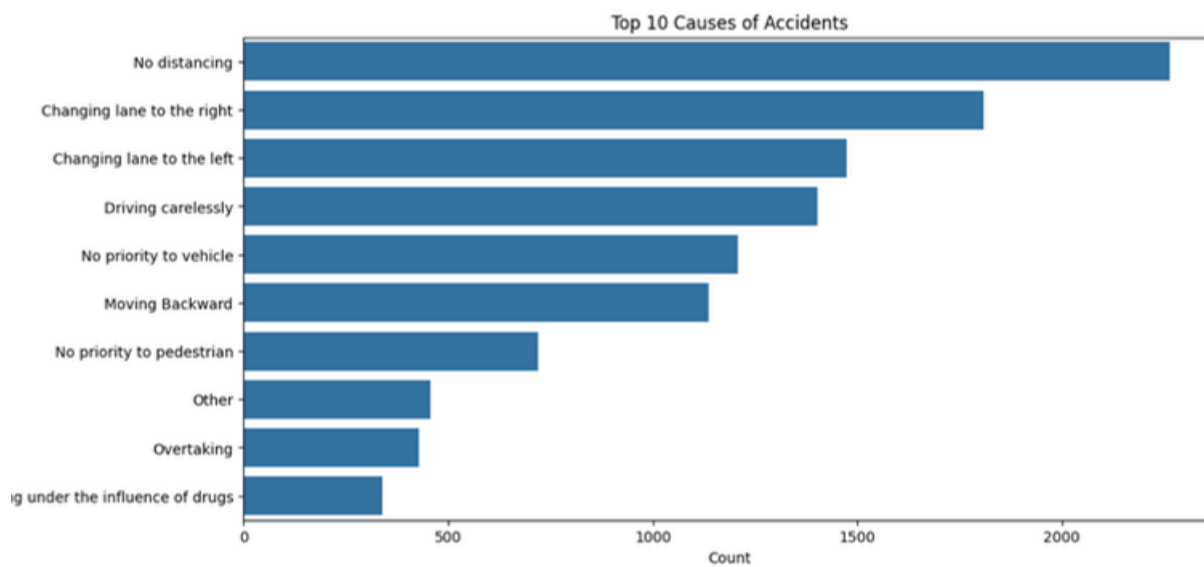
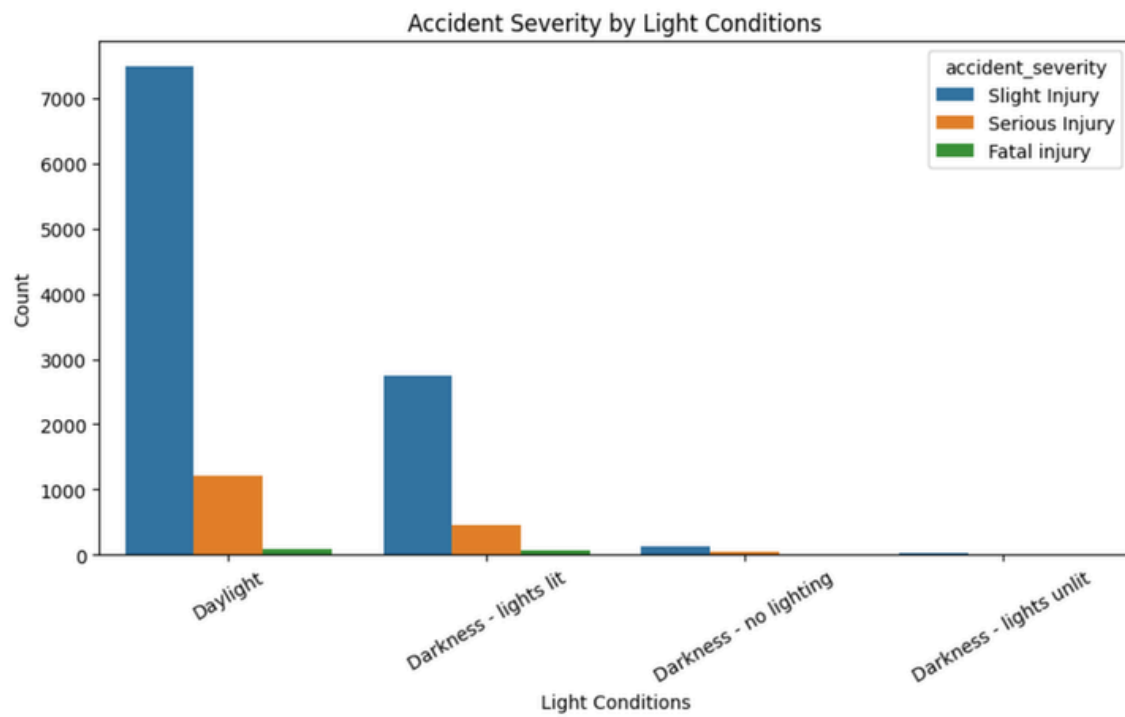
Data analysis is an important step in this project because it helps us understand the dataset before building the machine learning model. The Road Accident Severity dataset was analyzed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. The data was checked for missing values, cleaned, and prepared for further analysis.

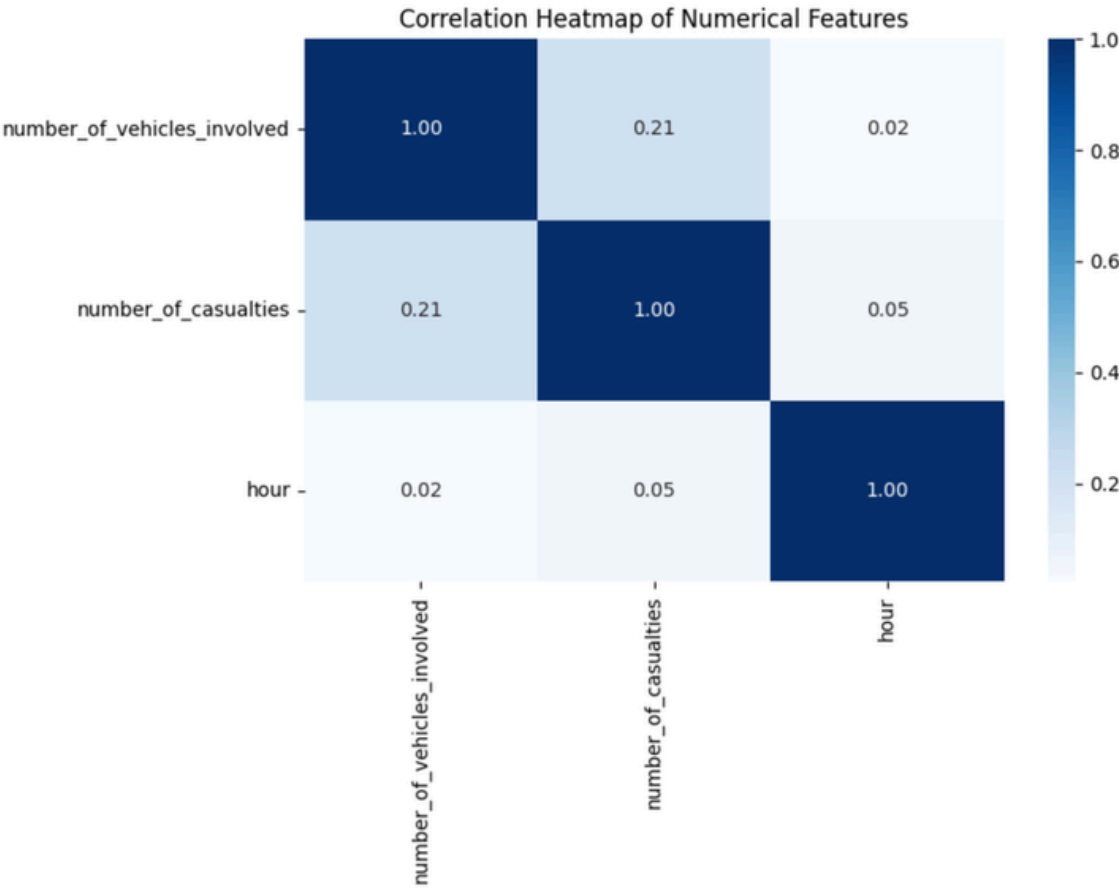
The dataset was first examined using functions like `head()`, `info()`, and `isnull().sum()`. Missing values in categorical columns were replaced with the most frequent value (mode). The Time column was used to create two new features: Hour and Time Period, which help improve the prediction accuracy.

After preprocessing, Exploratory Data Analysis (EDA) was performed using different charts and graphs to understand accident patterns based on factors such as weather conditions, driver age, road conditions, and accident severity. The insights gained from this analysis helped in selecting useful features for building the machine learning models.









7. MODEL DEVELOPMENT & RESULTS

Machine learning models were developed to predict the severity of road accidents based on different accident-related features. Before training the models, the dataset was divided into training and testing sets using an 80:20 ratio. The training data was used to build the models, while the testing data was used to evaluate their performance.

Since the dataset contained many categorical features, One-Hot Encoding was applied using a preprocessing pipeline to convert categorical values into numerical format. This ensured that the machine learning algorithms could process the data correctly.

Four classification algorithms were trained and evaluated:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)

The performance of each model was measured using Accuracy, Precision, Recall, and F1-Score.

Hyperparameter Tuning

After comparing all the models, Random Forest showed the best performance. To further improve its accuracy, GridSearchCV was used for hyperparameter tuning.

The following parameters were optimized:

- Number of Trees (n_estimators)
- Maximum Tree Depth (max_depth)
- Minimum Samples Split (min_samples_split)
- Minimum Samples per Leaf (min_samples_leaf)

Hyperparameter tuning helped improve the model's performance and select the best combination of parameters.

Machine Learning Models

Logistic Regression

Logistic Regression is a simple classification algorithm used to predict accident severity. It performs well on categorical data and provides a good baseline for comparing other machine learning models.

Decision Tree

Decision Tree classifies accident severity by creating decision rules based on different accident-related features. It is easy to understand and interpret but may produce lower accuracy due to overfitting.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. It reduces overfitting and provides more reliable results. In this project, Random Forest achieved the highest accuracy and was selected as the final model.

K-Nearest Neighbors (KNN)

K-Nearest Neighbors classifies accident severity by comparing a new record with its nearest neighboring data points. It is simple to implement and performs well on classification problems.

Model Comparison

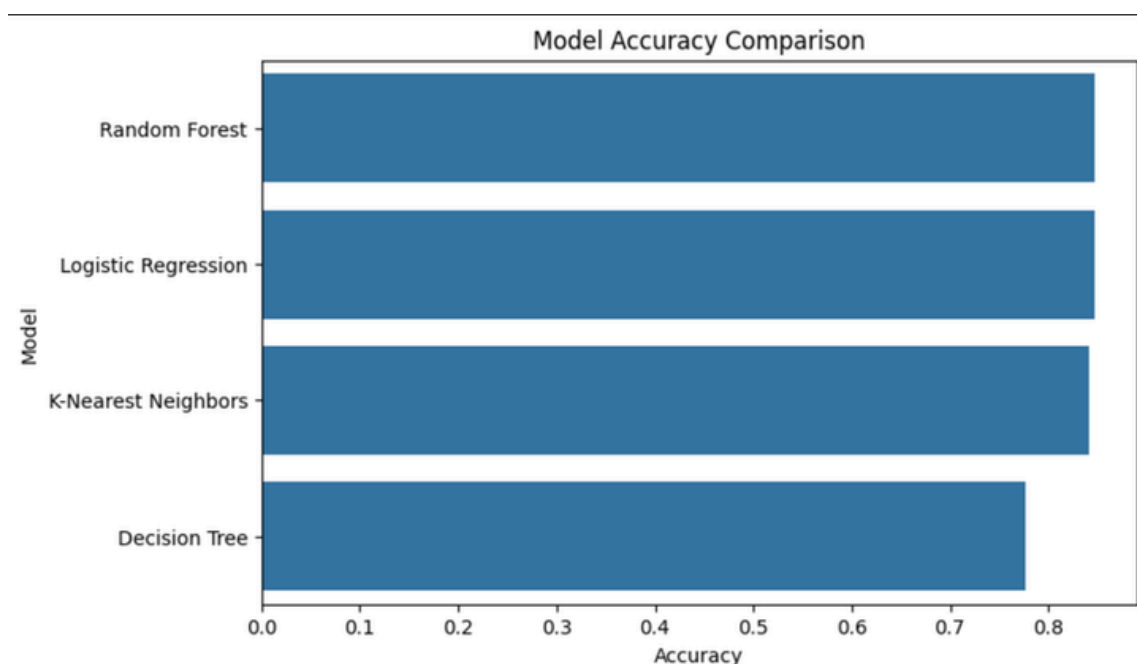
The performance of all machine learning models was compared using prediction accuracy. Among the four algorithms, Random Forest achieved the highest accuracy and was selected as the best model.

Hyperparameter Tuning

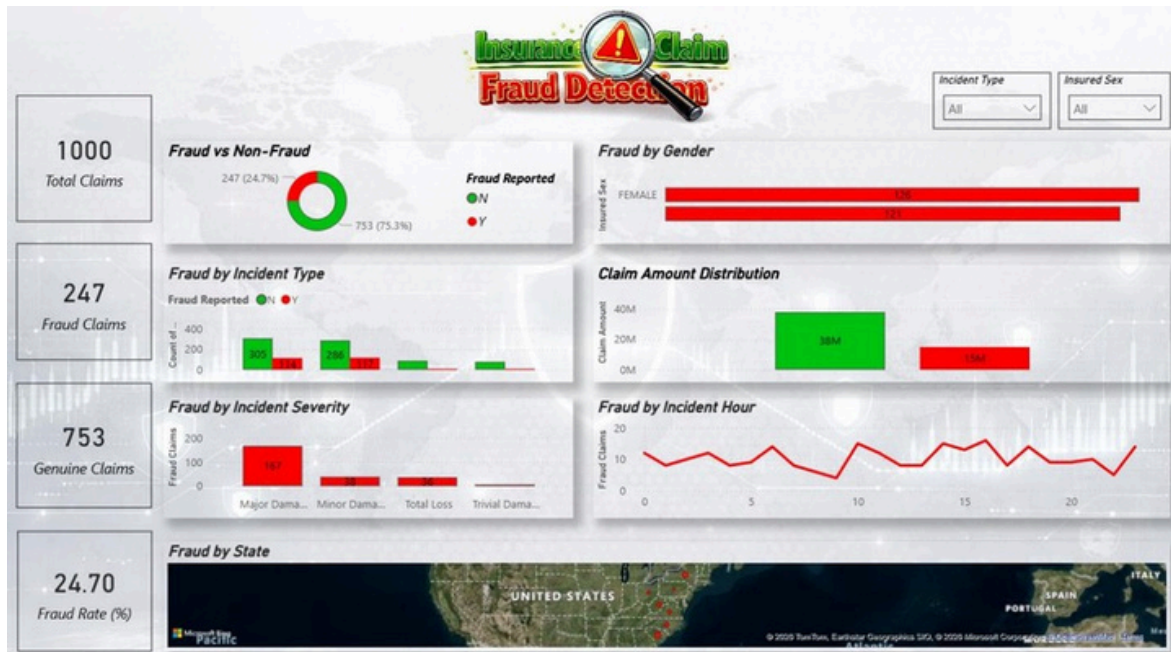
To improve the performance of the Random Forest model, GridSearchCV was used for hyperparameter tuning. Different combinations of parameters such as `n_estimators`, `max_depth`, `min_samples_split`, and `min_samples_leaf` were tested. The best parameter combination was selected automatically, resulting in improved model performance and better prediction accuracy.

Best Model Selection

Based on the comparison results, Random Forest achieved the highest accuracy (84.74%) among all the models. Therefore, it was selected as the final model and deployed in the Streamlit web application for accident severity prediction.



8. DASHBOARD OVERVIEW



The Power BI dashboard titled “Insurance Claim Fraud Detection” provides an interactive representation of fraud-related insights.

KPIs Displayed:

Total Claims

Total Fraud Claims

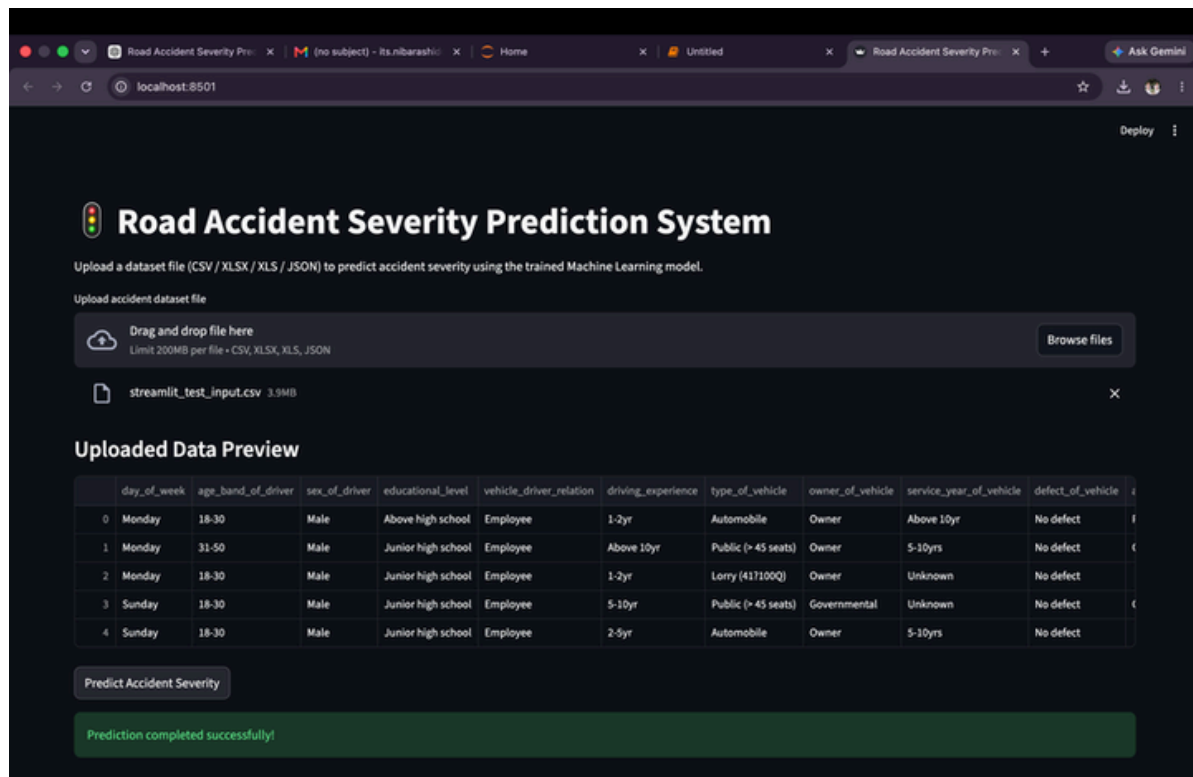
Total Genuine Claims

Fraud Rate

Visuals:

- Accident Severity Distribution (Donut Chart)
- Accidents by Driver Age Group (Stacked Bar Chart)
- Accidents by Weather Condition (Clustered Column Chart)
- Accidents by Day of the Week (Bar Chart)
- Accidents by Road Surface Condition (Stacked Column Chart)
- Accidents by Vehicle Type (Horizontal Bar Chart)
- Accidents by Cause of Accident (Clustered Bar Chart)
- Accidents by Time Period (Line Chart)
- Accidents by Light Condition (Pie Chart)
- Accidents by Number of Casualties (Column Chart)

9. STREAMLIT WEB APP DEPLOYMENT



Features of the Streamlit Application

- Upload accident data files (CSV, XLSX, XLS, JSON).
 - Preview the uploaded dataset.
 - Predict accident severity using the trained Random Forest model.
 - Display prediction results instantly.
 - Download the predicted dataset as a CSV file.
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10. CONCLUSION & FUTURE SCOPE

This project successfully developed a Road Accident Severity Prediction System using machine learning techniques. The dataset was cleaned, analyzed, and processed through data preprocessing and exploratory data analysis to identify important accident patterns.

Four machine learning algorithms—Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN)—were trained and compared. After hyperparameter tuning, the Random Forest model achieved the highest accuracy of 84.74% and was selected as the final model.

The model was deployed using Streamlit, allowing users to upload accident datasets and receive severity predictions. Additionally, a Tableau dashboard was created to visualize accident trends and provide meaningful insights. The project demonstrates how machine learning and data visualization can support road safety analysis and assist in decision-making.

Future Scope:

The Road Accident Severity Prediction System can be enhanced in several ways to improve its functionality and real-world usability.

Future improvements include:

- Integrating real-time traffic and weather data.
- Developing a mobile application for easy access.
- Using deep learning algorithms to improve prediction accuracy.
- Connecting the system with GPS and mapping services.
- Deploying the application on cloud platforms for public use.
- Expanding the dataset with recent accident records for better performance.

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